

BITS, PILANI – K. K. BIRLA GOA CAMPUS

Theory of Computation

(CS F51)

Lecture No. 5



Re Cap

Given an alphabet Σ , any set L such that

$L \subseteq \Sigma^*$ is called a **language over Σ**

Given an alphabet Σ and any words $x, y \in \Sigma^*$

We define informally a **concatenation** of words x, y as a word w obtained from x, y by writing the word x followed by the word y

We write the concatenation of words x, y as **$w = x \circ y$**

Definition

We define a **power** w^i of w by Mathematical Induction as follows

$$w^0 = e$$

$$w^{i+1} = w^i \circ w$$

Re Cap

Fact1:

For any alphabet Σ , any language over Σ is *countable*

Fact2:

For any alphabet $\Sigma \neq \phi$, there are *uncountably many* languages over Σ

Concatenation

Concatenation Definition

Given $L_1, L_2 \subseteq \Sigma^*$, a language

$$L_1 \circ L_2 = \{w \in \Sigma^* : w = xy \text{ for some } x \in L_1, y \in L_2\}$$

is called a **concatenation** of the languages L_1 and L_2

Concatenation can also be defined for languages on different domains

$$L_1 = \{w \in \{0, 1\}^* : w \text{ has an even number of 0's}\}$$

And $L_2 = \{w \in \{0, 1\}^* : w \text{ starts with 0 \& rest of the symbols (if present) are 1's}\}$

$$L_1 \circ L_2 = \{w \in \Sigma^* : w \text{ has an odd number of 0's}\}$$

Kleene Star

Kleene Star L^* of a language L is yet another operation specific to languages

We define L^* as the set of all strings obtained by concatenating zero or more strings from L

Remember that concatenation of zero strings is e , and concatenation of one string is the string itself

$$L^* = \{w_1 w_2 \dots w_k : k \geq 0, w_i \in L, i = 1, 2, \dots, k\}$$

Kleene Star

Kleene Star basic properties

$$e \in L^*, \text{ for all } L$$

$$L^* \neq \emptyset, \text{ for all } L$$

$$\emptyset^* \neq \emptyset$$

A natural question arises: is there any language $L \neq \Sigma$ such that nevertheless $L^* = \Sigma^*$?

Take $\Sigma = \{0, 1\}$ and take

$$L = \{w \in \Sigma^* : w \text{ has an unequal number of } 0 \text{ and } 1\}$$

$$L^* = \{0, 1\}^* = \Sigma^*$$

Kleene Star

$$L^+ = L \circ L^*$$

$$L^+ = \{w_1 w_2 \dots w_k : k \geq 1 \ w_i \in L, \ i = 1, 2, \dots, k\}$$

Show that

For some language L we have that $e \in L^+$ and

for some language L we can have that $e \notin L^+$

$$e \in L^+ \text{ iff } e \in L$$

Finite Representation of Languages

We can represent a finite language by **finite means** for example listing all its elements

Languages are often infinite and so a natural question arises if a **finite representation** is possible?

What do we mean by **finite representation**?

$L = \{w \in \{0, 1\}^* : w \text{ has } \mathbf{two} \text{ or } \mathbf{three} \text{ occurrences of } 1$
the **first** and the **second** of which **are not consecutive** }

Finite Representation of Languages

We will show that this language can be represented using only singleton sets and the symbols $\{, \}, (,), \cup, \phi, \circ$ and $*$

How the general string in language looks?

Questions?

1. Can the string begin with 0/1?
2. Can the string end with 0/1?

Etc. (will depend on the language description)

Finite Representation of Languages

The language can be represented as

$$L = \{0\}^*\{1\}\{0\}^*\{0\} \circ \{1\}\{0\}^*(\{1\}\{0\}^* \cup \emptyset^*)$$

We will dispense with the symbols braces and $\circ$

$$L = 0^*10^*010^*(10^* \cup \emptyset^*)$$

We call expression above (and others alike) a *regular expression*

Example

$$1^*01^* \cup (01)^*$$

Finite Representation of Languages

Language L on {0, 1} having strings of *even length*

Observe

Since 0 is even, $e \in L$

Any string of even length *can be obtained* by concatenating zero or more strings of length 2.

So $L = \{00, 01, 10, 11\}^*$

OR

$L = (00 \cup 01 \cup 10 \cup 11)^*$

OR

$L = ((0 \cup 1)(0 \cup 1))^*$

Regular Expression

Example: The Language L of Strings in $\{a, b\}^*$ with an Odd Number of a's

Note:

- A string with an odd number of a's has at least one a, and the additional a's can be grouped into pairs.
- There can be arbitrarily many b's before the first a, between any two consecutive a's, and after the last a

1st Attempt

$b^*ab^*(ab^*a)^*b^*$

What about ***$babababbaab$*** ?

Correct Answer

$b^*ab^*(ab^*ab^*)^*b^*$

Regular Expressions for identifiers in C

Example: An identifier in the C programming language is a string of length 1 or more that contains only letters, digits, and underscores (“_”) and does not begin with a digit.

If we use the abbreviations ***l*** for “letter,” either uppercase or lowercase, and ***d*** for “digit,”

then ***l*** stands for the regular expression

$a + b + c + \dots + z + A + B + \dots + Z$ ($\cup$ and $+$ means the same)

and ***d*** for the regular expression

$0 + 1 + 2 + \dots + 9$

The regular expression for the language of C identifiers is

$(l + _)(l + d + _)^*$